(3) Disconnect CMP electrical jumper harness (3) at engine wiring harness (Fig. 13).

(4) Before proceeding to next step, mark and note rotational position of oil pump drive housing (5) (Fig. 13) in relationship to engine block. After installation, the CMP sensor should face the 4 o'clock position as viewed from right side of engine.

(5) Remove hold-down bolt (4) and clamp (Fig. 13).

(6) Pull assembly from engine.

(7) Remove and discard old oil pump drive-to-engine block gasket.

INSTALLATION

2.4L

The Camshaft Position Sensor (CMP) on the 2.4L 4–cylinder engine is bolted to the right-front side of the cylinder head. **Sensor position (depth) is adjustable.**

(1) Remove plastic, upper timing belt cover (timing gear cover) (Fig. 15) by removing 3 bolts. Before attempting to remove cover, remove electrical connector from Engine Coolant Temperature (ECT) sensor (Fig. 15). This will prevent damage to sensor.

(2) Rotate (bump over) engine until camshaft timing gear and target wheel (tonewheel) are positioned and aligned to face of sensor as shown in (Fig. 16). **If not positioned as shown in (Fig. 16), damage to both sensor and target wheel will occur when attempting to start engine. Face of sensor MUST be behind target wheel while adjusting.**

(3) Position sensor to cylinder head and install 2 sensor mounting bolts finger tight.

(4) **SENSOR AIR GAP:.030"** Set air gap between rear of target wheel and face of sensor to.030". This can best be accomplished using an L-shaped, wire-type spark plug gapping gauge (Fig. 17). A piece of.030" brass shim stock may also be used.

(5) Gently push sensor forward until it contacts gapping gauge. **Do not push hard on sensor.** Tighten 2 sensor mounting bolts. Refer to torque specifications.

CAUTION: After tightening sensor mounting bolts, recheck air gap and adjust as necessary. Retorque bolts.

(6) Install upper timing belt cover and 3 bolts.

(7) Connect electrical connector to ECT sensor.

(8) Connect electrical connector to CMP sensor.

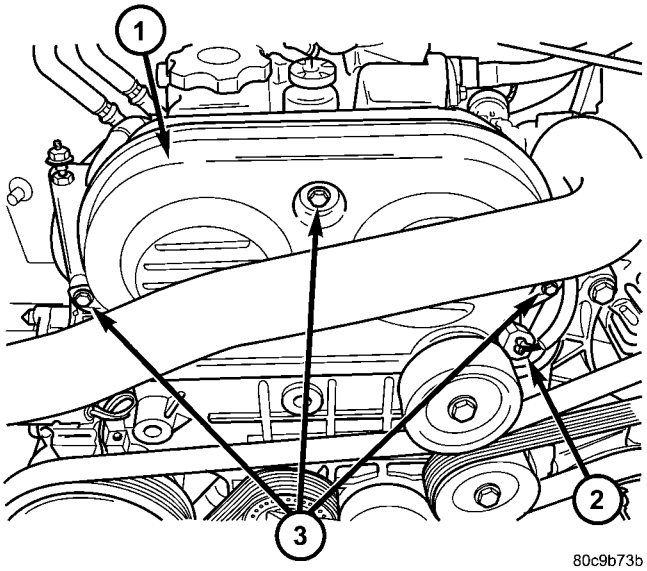

Fig. 15 UPPER TIMING BELT COVER/BOLTS-2.4L

1 - UPPER TIMING BELT COVER
2 - ELECTRICAL CONNECTOR (ECT)
3 - MOUNTING BOLTS (3)

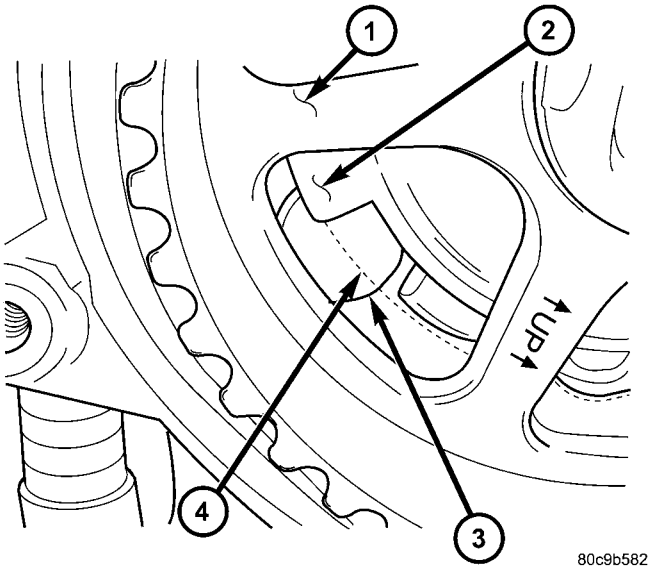

Fig. 16 CMP FACE AT TARGET WHEEL-2.4L

1 - CAMSHAFT DRIVE GEAR
2 - TARGETWHEEL (TONEWHEEL)
3 - FACE OF CMP SENSOR
4 - CUTOUT (NOTCH)